

Custom columns

INTERMEDIATE POWER QUERY IN EXCEL



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M Formula Language

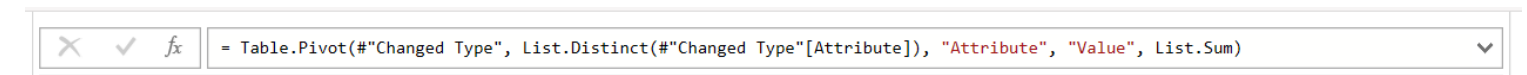
- 'M' for short (stands for Data Mashup)
- Functional programming language in Power Query
- Case-sensitive
- Wide range of built-in functions



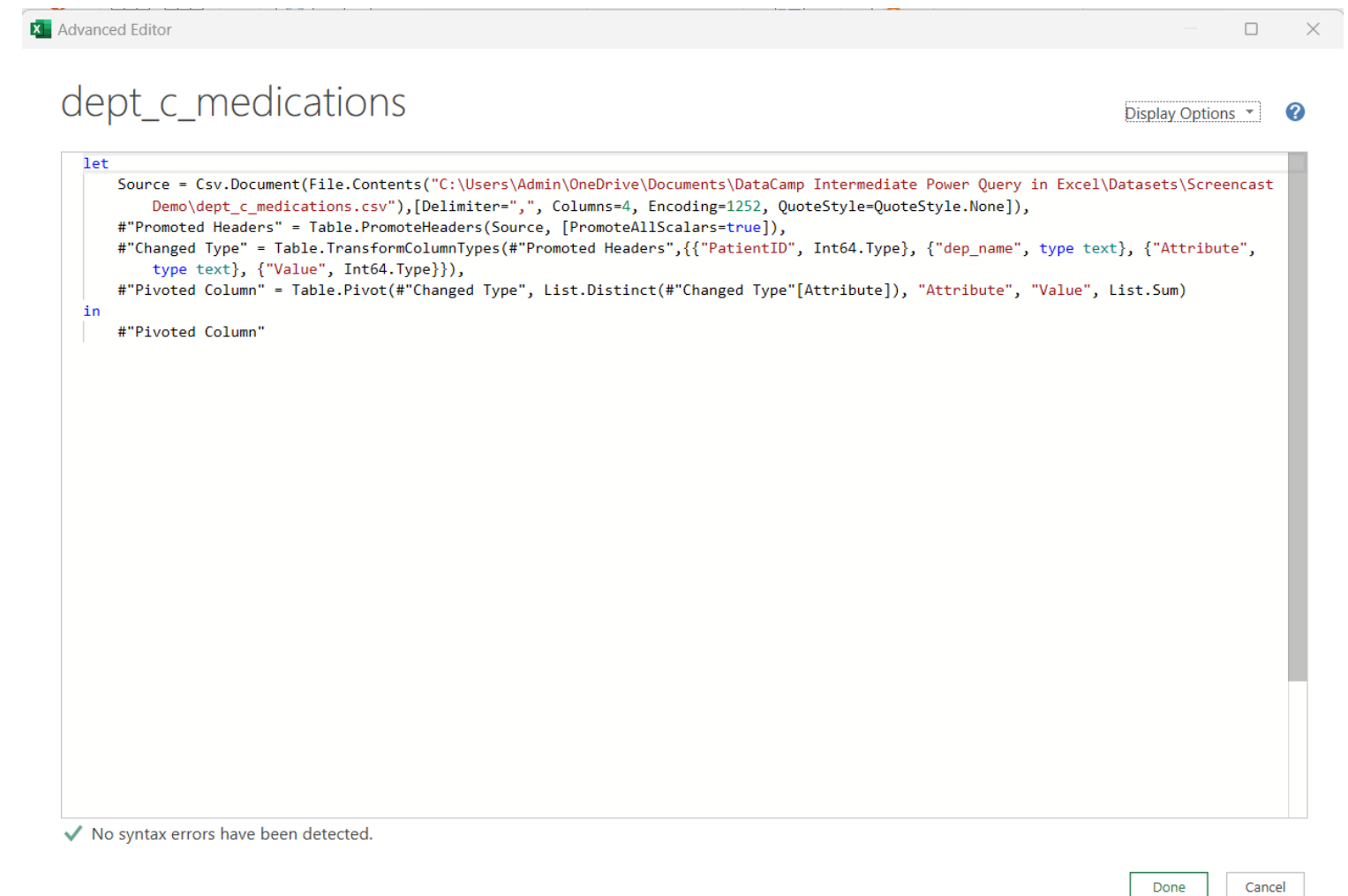
Generating M code

- Behind the scenes of every applied query step
 - Automatically written
- M code viewable:
 - Formula bar
 - Advanced Editor

Formula bar:

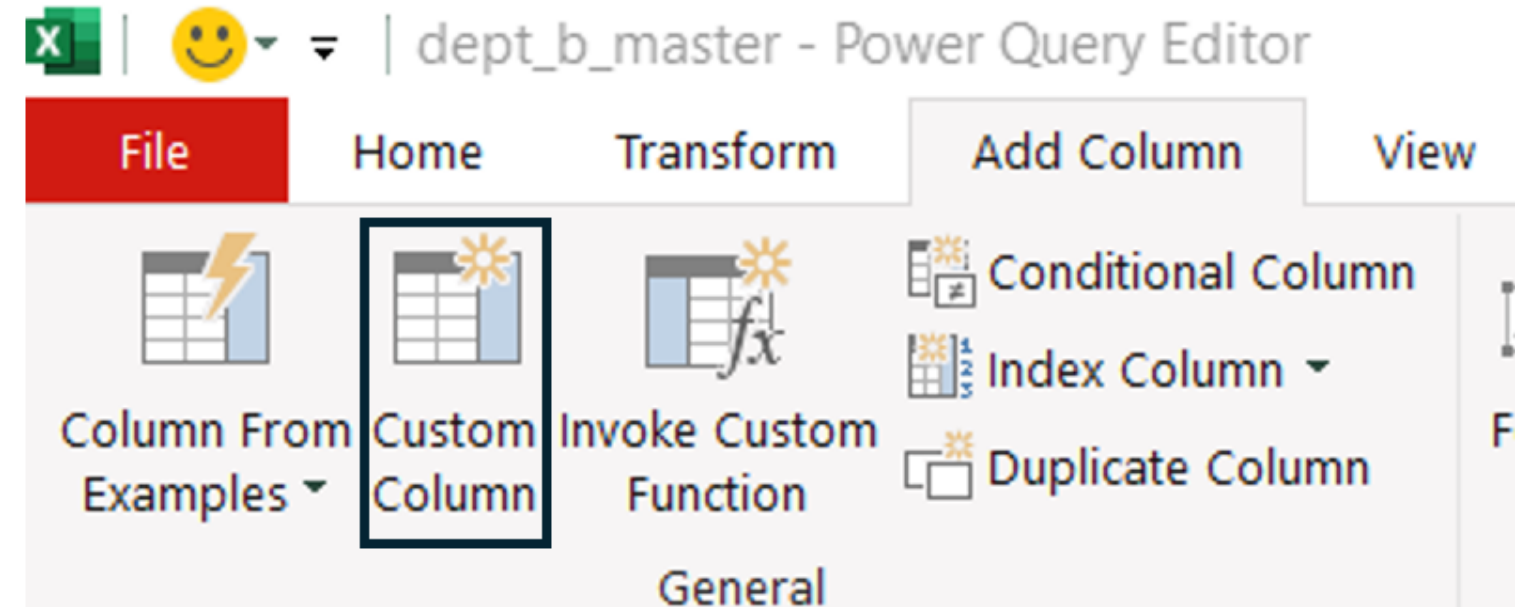


Advanced Editor:



Custom Columns

- User-defined calculated column
- Written in M language
- Can extend the functionality of built-in transformations.
 - Nested conditional logic
 - Advanced indexing
 - Complex calculations



Nested Conditional Logic

- Referencing columns and values
- Can involve multiple levels of conditional statements
 - If ... Then ... Else statements
- Can combine with logical operators
 - AND
 - OR

Basic Conditional Logic

```
if age >= 65
    and arrivalmode = "Car"
    then "group1"
else "group2"
```

Nested Conditional Logic

- Referencing columns and values
- Can involve multiple levels of conditional statements
 - If ... Then ... Else statements
- Can combine with logical operators
 - AND
 - OR

Custom Conditional Logic

```
if age >= 65 and age <= 80
    and arrivalmode = "Car"
then "group1"
    else if age >= 65 and age <= 80
        and arrivalmode = "ambulance"
    then "group1a"
else "group2"
```

Advanced Indexing



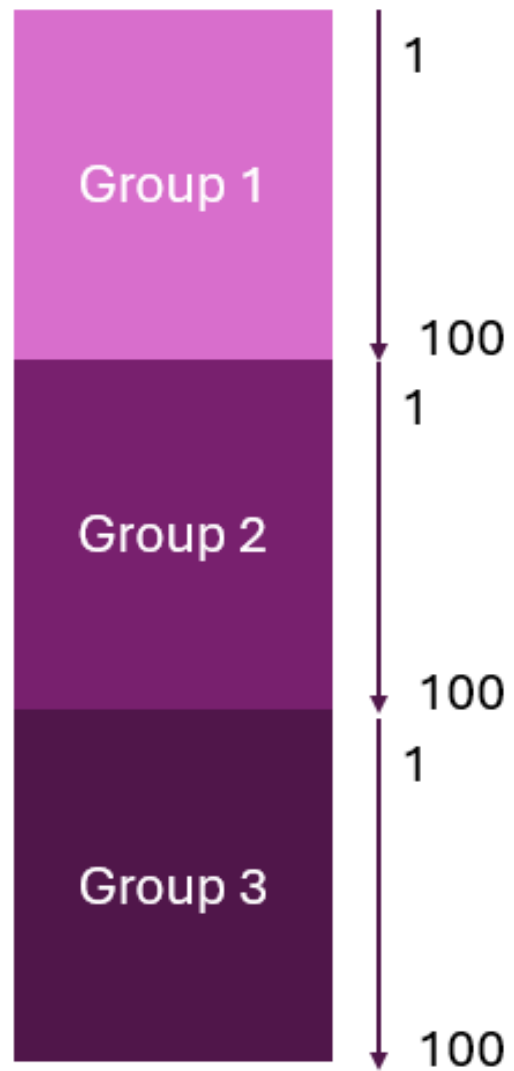
- Generate a custom index or rank based on defined groups

Advanced Indexing



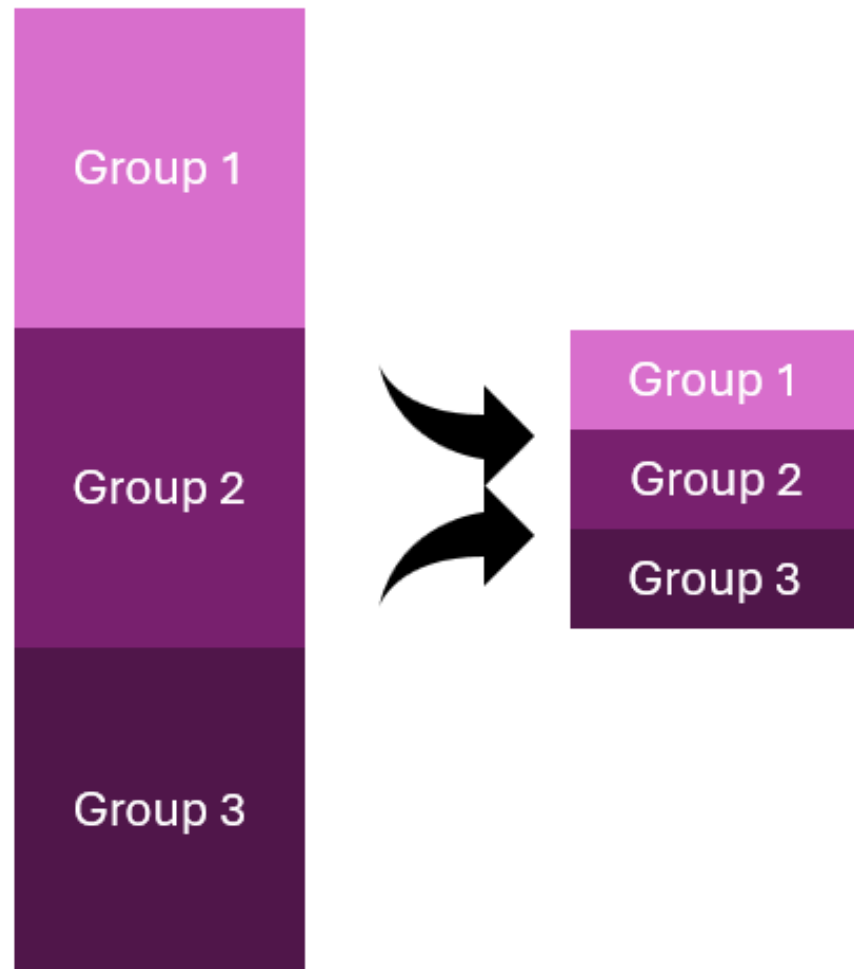
- Generate a custom index or rank based on defined groups
 - Simple Index

Advanced Indexing



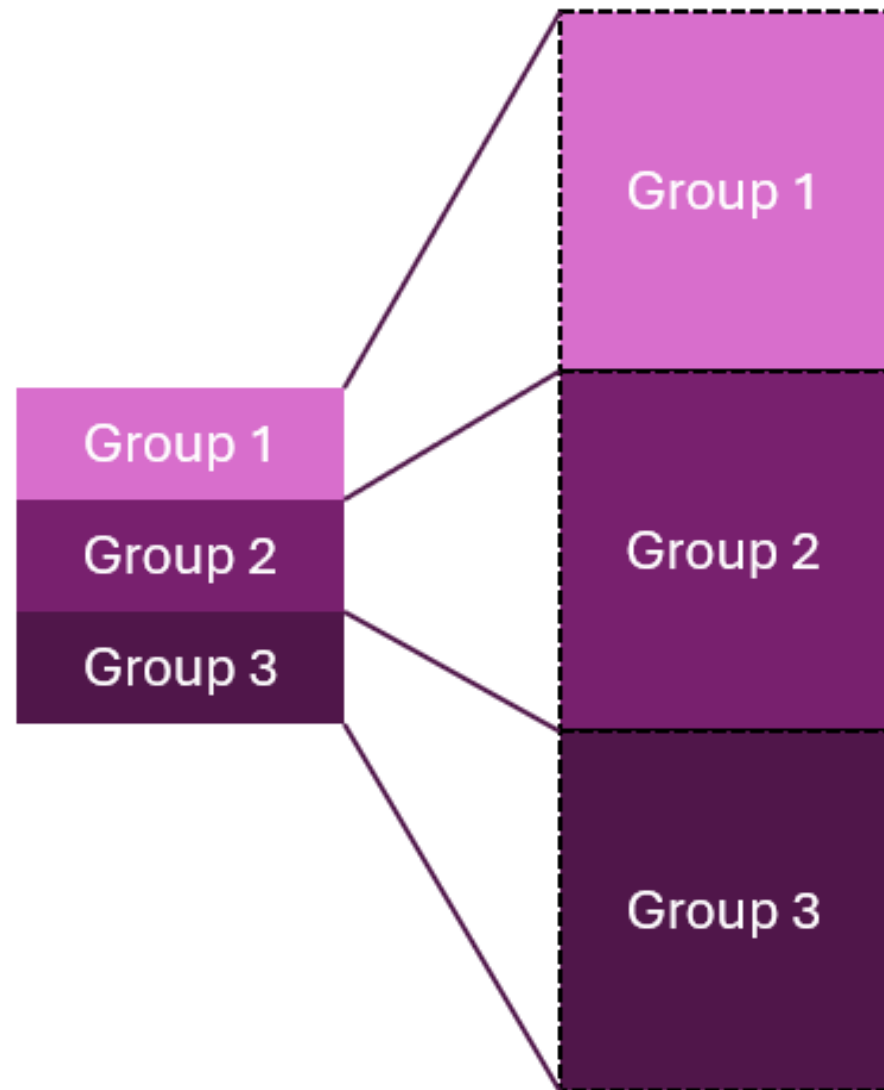
- Generate a custom index or rank based on defined groups
 - Advanced Index (within groups)

Advanced Indexing



- Generate a custom index or rank based on defined groups
- Group By operation with "All Rows" aggregation

Advanced Indexing



- Generate a custom index or rank based on defined groups
- Group By operation with "All Rows" aggregation
- Combine with custom M table function

```
Table.AddIndexColumn
```

Let's practice!

INTERMEDIATE POWER QUERY IN EXCEL

Custom columns in Power Query with M

INTERMEDIATE POWER QUERY IN EXCEL



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Let's practice!

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